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A comprehensive survey on intrusion detection systems with advances in machine learning, deep learning and emerging cybersecurity challenges

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Abstract
Recent advancements in technology, particularly in artificial intelligence (AI) and machine learning (ML), have significantly enhanced connectivity and efficiency in various domains. However, these developments have also led to increased threats and concerns regarding cyberattacks which have significantly challenged governments, businesses, and societies. An Intrusion Detection System (IDS) is a critical component of a cybersecurity architecture that is designed to monitor and analyze network traffic for signs of malicious activity and plays an essential role in detecting and responding to security threats. The increasing complexity of security attacks has led researchers to use machine learning (ML) and deep learning (DL) methods to enhance IDS performance. These cutting-edge systems are designed to detect emerging cyberthreats. The current study investigates and compares various types of IDS and AI-based techniques that use ML and DL algorithms to protect data against malicious threats. It also provides an overview of the architecture and operation of IDS and examines popular datasets such as CIC-IDS2017, KDDCup99 and UNSW-NB15 used to evaluate IDS models. These datasets offer a diverse range of attack scenarios and normal traffic and help create a more reliable and effective IDS which improves the overall network security. The study concludes by emphasizing constant progress in IDS technologies to combat evolving cyberthreats and enhance network security.

Keywords Intrusion detection system, Deep learning, Machine learning, S-IDS, A-IDS, NIDS, HIDS, IDS architecture

1 Introduction

Adapting to growing populations and managing and addressing the associated challenges in urban areas require intelligent infrastructure [1]. Our world faces a surge in the number of network devices due to the technology advancements, the increased connectivity, and integration of intelligent solutions in various fields. fundamental technologies such as Internet, cloud computing, Internet of things (IoT), big data, and modern

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industrial control systems has significantly improved in recent years. They are often coordinated with each other to manage vast amounts of data over the network. Such synergy is crucial in managing the information flow that modern societies rely on [2].

Today's modern world deeply relies on complex software systems used in various industries ranging from healthcare and financial to education and transportation. This reliance makes such systems primary targets for cyberattacks. The threats posed on the businesses, citizens, and societies as a whole is a sad reality of life in the modern world [3–5]. The ever-expanding list of threats includes malwares, worms, trojans, spywares, ransomwares, drive-by compromises, man-in-the-middle attacks, denial-of-service attacks, various types of intrusion attacks and wide range of other malicious behaviors reaching even advanced data hiding techniques. The spread and emergence of advanced networking technologies is a transformative force in the modern societies. These technologies have brought unprecedented comfort, a significant increase in efficiency, simplified daily activities, and changed our lifestyles. Unfortunately, they also expand the area under attack and bring cyberthreats closer to the homes of ordinary citizens and critical industries [6–8]. The sensitivity of critical infrastructures enables these attacks to cause cascading effects. The advancement of cyberthreats turns protecting sensitive data and privacy into a major challenge. The consequences of security events can range from financial losses to major disruptions in public and private services, making the stakes incredibly high [9] Thus, the effort to strengthen security measures are increasingly critical than ever. This requires a multifaceted approach that includes developing cybersecurity protocols, constant monitoring, updating network defense systems, and fostering a culture of security awareness among users [10].

Conventional security mechanisms, such as firewalls and antivirus software, are increasingly challenged by attackers adept at creating novel and sophisticated attacks. Conventional tools hardly keep up with the attackers smart tactics. Advances in mobile communication technology have also facilitated the sharing of a wide range of varying and heterogeneous data all over the network. The enhanced and extended connectivity has come with fresh vulnerabilities and weaknesses. The large number of access points in these systems makes them vulnerable to sophisticated cyberattacks. Facing these challenges requires novel security solutions that keep up with changing cyberattacks. Networking technologies have become an essential part of our life, highlighting the need to improve security measures against ever-changing cyberthreats [11–14].

1.1 Intrusion detection system: increasing cybersecurity through advanced technologies

Cybersecurity consists of strategies to protect data, applications, servers, and the network against unauthorized access or modification and ensuring the integrity, confidentiality, and availability of data. Technological advancements increases cyberthreats, making data security an essential priority for businesses, governments, and individuals [15, 16]. Network security, a core part of cybersecurity, focuses on defending networked systems against malicious activity. Development of intelligent systems that adapt to evolving threats and ensuring users safety holds the key to the future of cybersecurity [16]. Modern cybersecurity research heavily invests on IDSs. They are key components in the detection and mitigation of anomalous activities. These cutting-edge systems are designed to detect emerging cyberthreats. Their goals are accurate threat detection, reducing false alarms, and maintaining a robust defense against various types of

cyberthreats [17–19] and provide a fundamental solution to security and application monitoring issues. An IDS is designed to detect intrusions and potential network or system abuse, send prompt alarms to managers, and document and log events for future analysis. Despite their usefulness, conventional IDSs face limitations and drawbacks [6]. An intrusion prevention system (IPS) enhances security by preventing attacks. It examines network traffic patterns to find behaviors that deviate from established norms and logged data history. This approach seeks to prevent a security breach before it causes any damage [20].

An IDS is usually considered the first line of defense against threats, provides systems and network protection by detecting unauthorized activities [7]. Intrusion Detection Systems (IDS) are generally classified into Network-based (NIDS) and Host-based (HIDS). While NIDS monitors network traffic for suspicious patterns, HIDS focuses on events and activities within individual hosts (Fig. 1). They fall into three main categories [2]:

1.1.1 Signature-based IDS (S-IDS)

These systems rely on a database of known attack patterns or signatures. Signatures act as cyberattacks' fingerprints. The IDS examines network traffic, system reports, and files for these specific patterns. Once a match is found, it generates an alert.

1.1.2 Anomaly-based IDS (A-IDS)

An anomaly-based IDS is designed to detect unknown or new attacks. It establishes a baseline for normal network or system behavior. Then it continuously monitors all the activities, looking for any behavior that deviates from that set standard.

1.1.3 Hybrid IDS (H-IDS)

1.1.4 These systems combine the strengths of S-IDS and A-IDS to provide a more comprehensive defense.

1.2 Key factors in the accuracy and timely detection of attacks by IDS technology are [21]

- System stability and reliability
- Rapid detection
- Minimum false positives
- Maximum detection rate and coverage of most types of threats
- Minimal use of software/hardware
- Ability to accurately detect the location of penetration
- Ability to work and integrate with other technologies

1.3 Motivation

The sophisticated defense offered by a modern IDS against the complex and evolving threats makes it a crucial part of cybersecurity technology. Advanced technologies such as machine learning (ML) and deep learning (DL) are integrated into such systems to provide advanced adaptability, comprehensive monitoring, and the ability to detect and mitigate zero-day attacks. This means faster adaptation to new threats and more efficient protection of sensitive data and services. Such systems enhance systems security

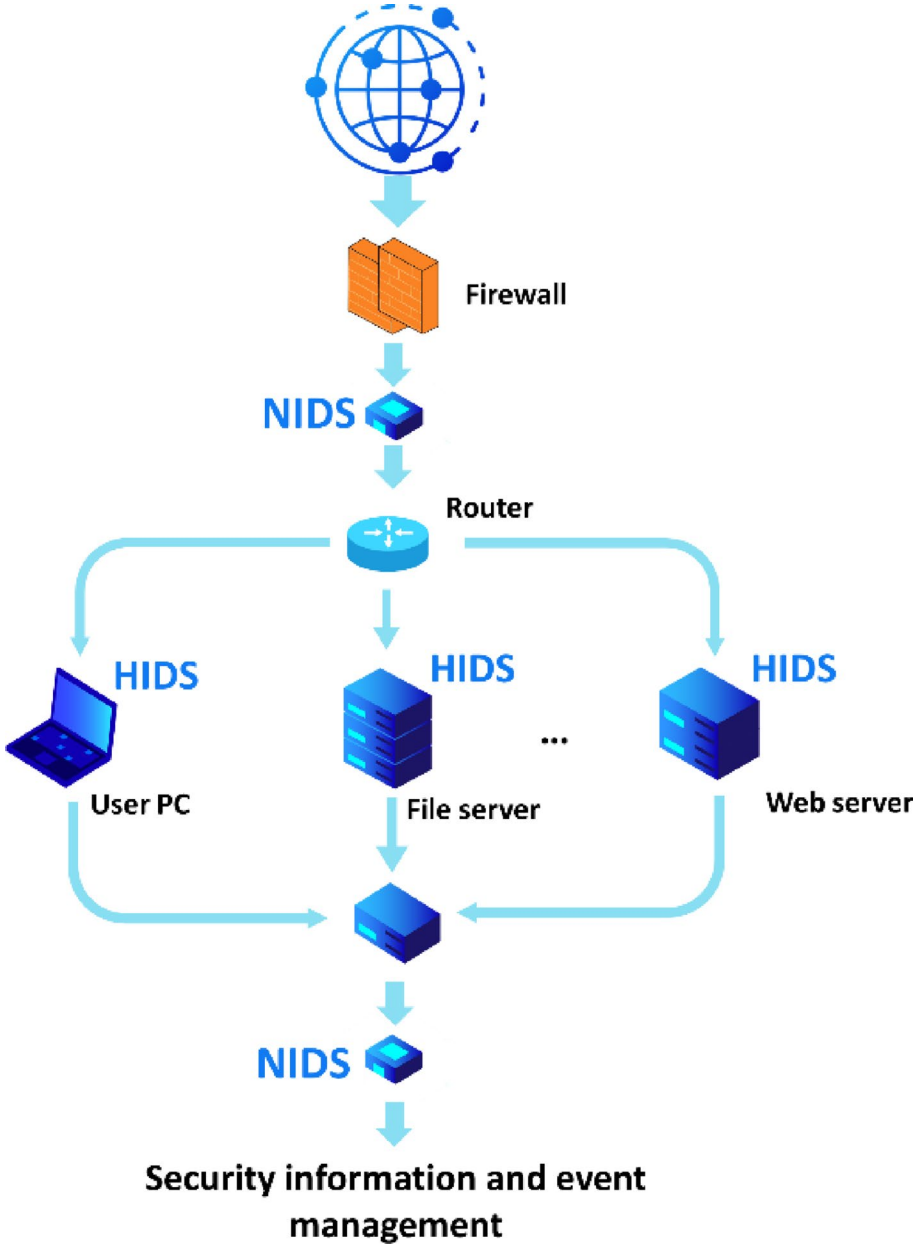

Fig. 1 Basic architecture and placement of NIDS and HIDS

by adding an additional layer of defense beyond conventional firewalls. Advanced IDS solutions also play a crucial role in preventing major threats, protecting vital data, and ensuring sustainability of services which makes them a necessary evolution in fighting cybercrimes.

1.4 Related works

Studying IDSs have attracted significant attention in the recent years due to the increasing complexity and frequency of cyberthreats. Numerous researchers have contributed to it by investigating various methods and technologies that enhance the effectiveness of IDS solutions.

Hodo et al. [22] presented a comprehensive classification and review of IDSs using shallow and deep networks. They examined various ML techniques and their performance in anomaly detection, emphasizing the role of feature selection in the classification and training stages of the IDS. The study also discussed challenges of creating reliable and efficient IDS solutions capable of handling large amounts of data and adapt to changes in real-time. The issues of false positive and false negative rates and providing insight into modeling IDS solutions based on ML are also discussed [22].

Liu and Lang [23] provided an extensive survey on the application of ML and DL techniques in IDS solutions. They classified IDS studies according to their data objects and summarized different ML and DL techniques applied to anomaly Detection. The study discussed the strength and limitations of those methods and highlights their ability in improving the detection precision and large-scale data management. Key challenges such as high false alarm rates and the need for real-time detections were also discussed. The survey is a valuable resource in understanding the current advances and challenges in using ML and DL in IDS [23].

Aldweesh et al. [24] gave a comprehensive study of the DL approaches in anomaly-based IDS. They precisely classify current FL-based IDS solutions and cover various aspects such as input data, detection methods, deployment strategies, and evaluation techniques. The study highlights the efficacy of DL in managing big data and improving detection precision. Challenges and open issues are also discussed and the need for further research to improve the efficiency and strength of IDS is emphasized. The survey is a valuable resource in understanding the current advances and challenges in using ML and DL in IDS [24].

Lansky et al. [25] present a systematic survey of using DL techniques in IDS. They provide a comprehensive overview of the DL methods such as autoencoders, recurrent neural networks (RNNs), and convolutional neural networks (CNNs) used to improve the IDS performance. The study categorizes the methods based on their architecture and discusses their effectiveness in network intrusion detection. It also highlights future challenges and paths in the field and the need for stronger and more adaptable IDS [25].

Halbouni et al. [6] offer a general overview of using ML and DL techniques in cybersecurity. The authors examine various ML and DL algorithms used for intrusion detection, malware detection, and other cybersecurity applications. They highlight the efficiency of those approaches in detecting and mitigating cyberthreats. Incoming challenges in executing ML and DL techniques in cybersecurity are also examined and future research directions for enhancing the strength and adaptability of such systems are proposed [6].

Pawlicki et al. [8] offer a comprehensive survey of using neural networks (NNs) in cybersecurity and security challenges of the NNs themselves. Table 1 They discuss different architectures for NNs and their use in detecting and mitigating cyberthreats. The study furthermore emphasizes the vulnerabilities of NNs to adversarial attacks and other security issues [8].

Alkasassbeh et al. [26] present an advanced survey of IDS. The authors provide a comprehensive taxonomy of IDSs, categorizing them into signature-based, anomaly based, and hybrid ones. The features and efficiency of each category is discussed and the challenges and current gaps in the current IDS technologies are highlighted [26].

The study gives an overall view of current IDS research and highlights significant scientific advances in the field. Valuable insight is offered regarding different methods,

Table 1 Article abbreviations

IDS intrusion detection system	IPS intrusion prevention system
S-IDS signature-based IDS	A-IDS anomaly-based IDS
H-IDS Hybrid IDS	AI Artificial Intelligence
ML Machine learning	DL Deep learning
LR Logistic regression	KNN K-Nearest neighbors
PCA Principal component analysis	DT Decision trees
RF Random forest	SVM Support vector machine
EO Equilibrium optimization	ASO Atom search optimization
Bagging bootstrap aggregating	AdaBoost Adaptive boosting
XGboost Extreme gradient boosting	CNN Convolutional neural networks
MLP Multi-layer perceptron	DBN Deep belief networks
RBM Restricted boltzmann machine	RNN Recurrent neural networks
LSTM Long short-term memory	AE Autoencoder
APT Advanced persistent threats	CIC Canadian cybersecurity institute
CAN Controller area network	DoS Denial of service
CSE Communications security establishment	DDoS Distributed DoS
ECU Electronic control unit	IoV Internet of vehicles
IoT Internet of things	NN Neural networks
NIDS Network based IDS	HIDS Host based IDS
GNNs Graph neural networks	AML Adversarial machine learning
FL Federated learning	XAI Explainable AI

technologies, and approaches that are applied to increase the efficiency of IDS. These include the application of ML and DL techniques, the integration of artificial intelligence in IDS, and the development of new algorithms and models for anomaly detection. Continuous efforts in this field play a vital role in developing strong and adapting IDS solutions capable of protecting digital infrastructures against evolving cyberthreats. The increasing complexity of cyberthreats requires ongoing research and development to ensure effective detection and mitigation of the potential intrusions which protect sensitive data and preserve the integrity of digital systems. This study builds upon existing literature by providing an integrated overview of IDS architectures, widely used machine learning and deep learning models, and benchmark datasets. Although it does not propose a new algorithm or framework, it aims to consolidate key elements from previous works to offer a clearer understanding of current research directions and challenges in the IDS field. Table 2 provides a comparative analysis of recent studies on ML- and DL-based Intrusion Detection Systems (IDS). In contrast to prior works, such as Hodo et al. [22], which emphasized feature selection, or Liu and Lang [23], which focused on categorizing ML and DL techniques, this study consolidates an overview of IDS architectures (S-IDS, A-IDS, H-IDS) alongside modern benchmark datasets, including CICIDS2017 and UNSW-NB15. This synthesis highlights key research gaps, such as the need for scalable and adaptable IDS solutions to address evolving cyber threats.

2 Comparing IDS solutions

IDS solutions, an essential part of network security, are designed to detect unauthorized accesses or malicious activities in a network. Evolving malware and increasingly complex attacks are a key challenge to IDS. Zero-day attacks are a major concern which highlights the need for effective IDS. IDS solutions are categorized into two main categories: signature-based and anomaly-based [16, 20] (Fig. 2).

Table 2 Comparative analysis of recent studies on machine learning- and deep learning-based intrusion detection systems

Study	Year	Main focus	Techniques	Dataset(s)	Key contributions	Limitations
[22]	2017	Shallow & deep networks in IDS	ML models, feature selection	Not specified	Classification of IDS types; ML feature role	Data volume challenge, false positives
[23]	2019	ML/DL for anomaly detection	SVM, CNN, RNN, LSTM	NSL-KDD, KDD99	Categorization of ML/DL methods, detection precision	High false alarms, real-time infeasibility
[27]	2019	Android malware prediction using ELM	ELM (poly, linear, RBF kernels), other classifiers, sampling (upscale, etc.), feature selection (Wilcoxon, Gini)	1600 APKs (1000 benign from Google Play and other stores, 600 malware from Malware Genome and DroidKin)	- Poly kernel ELM best (76.6% accuracy) - Upscale sampling (79.2% accuracy) - Wilcoxon feature selection effective	- Imbalanced data - High-dimensional data - Complex aggregation
[24]	2020	DL in anomaly-based IDS	DBN, CNN, autoencoders	CIC-IDS2017	DL taxonomy, deployment strategies	Scalability, training cost
[25]	2021	DL architectures in IDS	CNN, RNN, AE	UNSW-NB15, custom	Architecture-based classification, future directions	Adaptability challenges
[6]	2022	ML/DL in cybersecurity	Various ML/DL models	Multiple	General overview of techniques, future research needs	Implementation challenges
[8]	2022	NNs in cybersecurity	Various NN types	Not specified	Security issues in NNs, adversarial vulnerabilities	Adversarial attacks threat
[26]	2023	IDS taxonomy and classification	ML & DL, hybrid IDS	CIC-IDS2017, UNSW-NB15	Comprehensive taxonomy of IDSs	Highlighted gaps, limited benchmarking
[28]	2025	XAI overview	LIME, SHAP, CFE, Saliency Maps, LRP, CAM, Integrated Gradients, Attention, TSViz	Unspecified (NLP, vision, time series, healthcare)	Comprehensive review of XAI needs, techniques, applications, and future directions; taxonomy of methods; covers 200 + studies	Lacks quantitative comparisons; limited real-world implementation details; shallow ethical/legal discussion

2.1 S-IDS

Signature-based detection looks for signs of predefined patterns which enables the detection of certain patterns in the network traffic. It relies on predefined patterns to detect specific threats in network traffic. This is achieved by comparing the observed data to a pattern database and signatures of predefined threats. Once a match is found, the system generates an alert. Ignoring new and emerging threats due to the lack of their signature is the main shortcoming of this category [16, 19, 29, 30].

- Signatures are unique patterns or identifiers associated with certain types of attacks or malwares.
- Signature-database: S-IDS maintains a comprehensive database of known threats' signatures, which can include byte sequences in network traffic, sequences of known malicious instructions used by malware, or patterns indicative of known attack strategies.

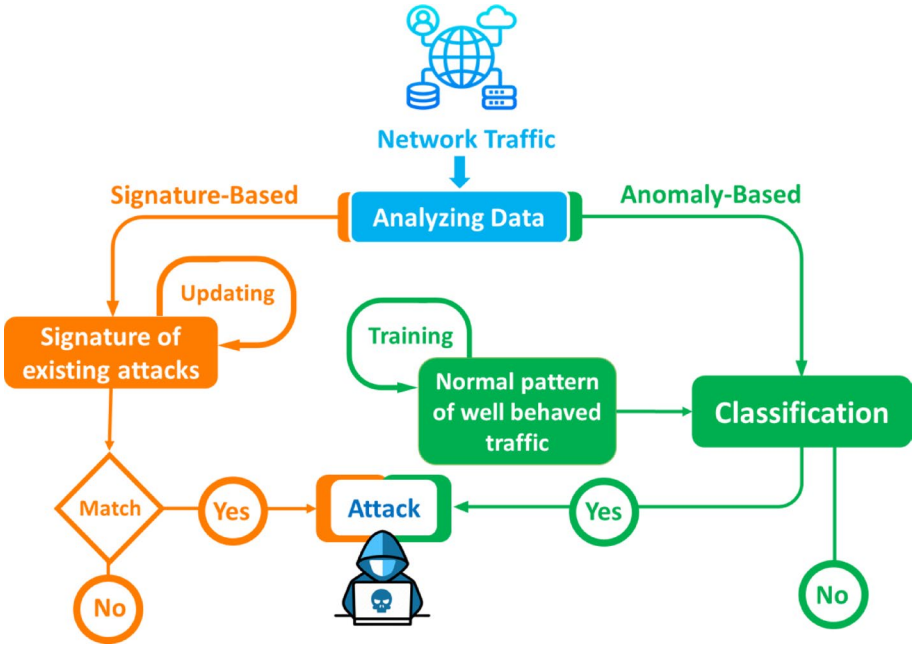

Fig. 2 Signature- and anomaly-based IDS

2.2 A-IDS

Anomaly detection, also known as outlier detection, is a data analysis technique that detects data points and events that deviate from the baseline or normal behavior of data. Anomalies stand out from the majority of the data and deviate from expected or typical patterns. However, detecting them includes challenges such as: unlabeled real-world data, the rarity of anomalies compared to normal data, adaptive behavior of anomalies, and false positive detections [16, 19, 31].

Anomaly detection has crucial applications in various fields of today's modern life such as network security (intrusion detection and abnormal behavior detection), health monitoring (detecting abnormal patient conditions), financial fraud detection (tagging suspicious transactions), industrial systems (predicting equipment failure), environmental monitoring (detecting sings of pollution).

In the realm of network security, anomaly detection plays a crucial role in safeguarding network infrastructure, ensuring operational stability, and enabling organization to swiftly detect and handle security threats to ensure that networks remain secure, data integrity is maintained, and legal obligations are met. It is an essential tool in the modern world of cybersecurity.

Each of S-IDS and A-IDS solutions has their advantages and disadvantages (Table 3) and are often used together to provide a more comprehensive security. S-IDS is more reliable in detecting known threats while A-IDS has the adaptability required to respond to modern and emerging cyberthreats. Using them together provides a powerful defense.

mechanism against a wide range of networks intrusions that enhances detection capabilities and reduces false positives [2, 32].

3 The architecture of an IDS

IDS solutions consists of several key components that work together to monitor, analyses, and respond to potential security threats. These components are (Fig. 3):

Table 3 A comparison of signature- and anomaly-based IDS

Aspect	S-IDS	A-IDS
Detection approach	Known patterns (exact matches)	Deviations from normal behavior
Speed	Higher speed due to predefined list of signatures	Can detect unknown or novel attacks without relying on predefined signatures
Accuracy	High accuracy for known attacks	focuses on identifying deviations from normal behavior
Coverage	Limited to known threats	Broader coverage, including zero-day threats
Adaptability	Static (requires updates)	Dynamic (adapts to new threats)
False positives	Low	Higher
Cases	Well-suited for common attacks	Effective against novel threats
Hybrid approaches	Some IDS combine both methods	

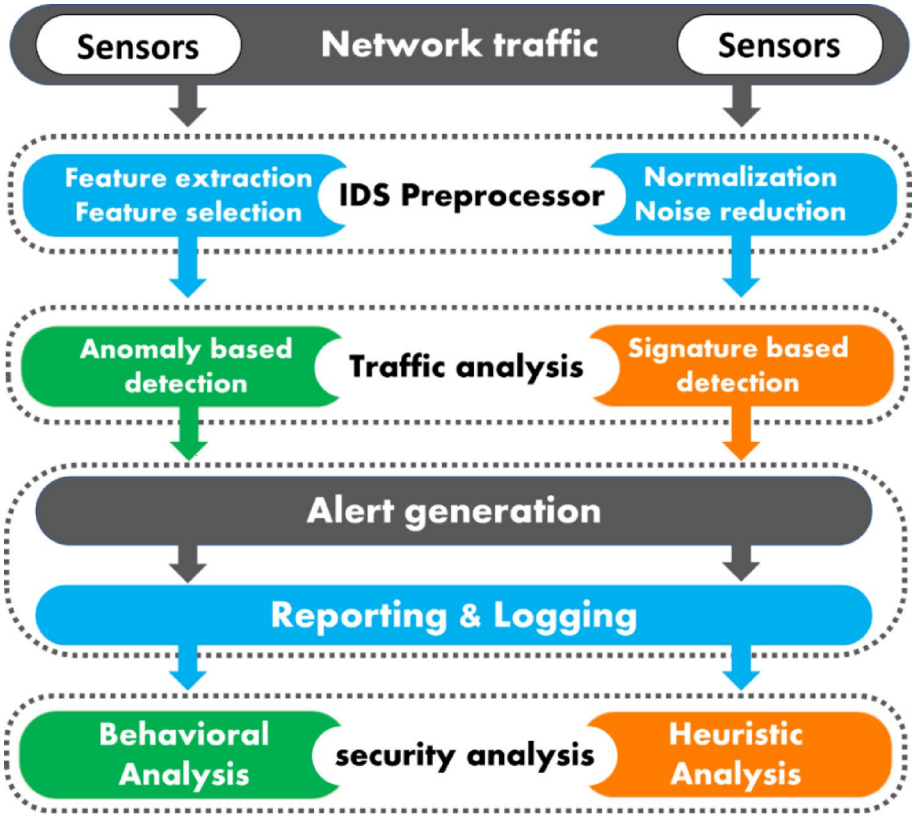

Fig. 3 The Architecture of an IDS

3.1 Network sensors

Network sensors actively monitor the flow of packets and data across the network, including inbound and outbound traffic. They gather the raw information of packets as they pass through the network. The raw data captured is sent to the preprocessor. The preprocessor normalizes data and prepares it for further analysis by detection engines. Simply put, network sensors are watchdogs of the network that collect network data and transmits it for further processing [33, 34].

3.2 Preprocessor

Raw data usually includes noise, and lost, redundant, or missing values. Preprocessing prepares the raw network data for further analysis. The processed data is then

normalized and becomes standard and suitable for future processing [33]. Key properties such as source and destination IP address, port number, and other pertinent data are identified and extracted from the data at this stage. Extracting these features allows subsequent detection engines to focus on specific aspects of network traffic.

The preprocessed data is then fed to relevant detection engines (such as anomaly detection and signature matching ones). Those engines use that preprocessed data to detect potential threats or anomalies in the network [21].

3.3 Traffic analysis: detection engines

3.3.1 Anomaly detection engine

This engine learns the baseline normal behavior of the network and system. Once a deviation from the baseline (anomaly) occurs, the model generates an alert. Such engines are quite helpful in detecting emerging and unknown attacks [34, 35].

3.3.2 Signature matching engine

These engines compares network traffic to a database of known attack patterns (signatures) and generates an alert if a match is found. These engines help detecting known attacks with consistent signatures and patterns [34, 35].

3.4 Alert generator

The alert generator collects alerts from anomaly- and signature matching engines and prioritizes them based on their severity. Alerts are formatted for human analysis. The alert generator also logs the relevant information of detected events. These include details such as timestamps, source/destination IP addresses, port number, and the nature of the alert (e.g., type of attack detected). In summary, alert generator plays a crucial role in reporting security incidents by generating alerts and maintaining logs for later analysis [21, 34].

3.5 Security analysts

The security analysts review and evaluate the alerts generated by the IDS to find false positives or false negatives. Their analysis is the basis of the appropriate decisions and measures, such as blocking the network traffic [19, 21].

4 Techniques used in IDS solutions

Early IDS solutions relied on rule-based approaches. These systems used predefined signatures, or patterns, to identify known attacks. They lacked adaptability and struggled with new and emerging threats. Reliance on rigid rules and constant need of manual update made them unable to handle zero-day attacks or emerging attack techniques [20, 36, 37]. Researchers introduced AI techniques to enhance IDS performance. AI brought adaptability, scalability, and accuracy to IDS. AI models adapt to emerging threats by learning and adapting over time [37, 38].

4.1 Machine learning

ML is a subset of artificial intelligence. It focuses on algorithms that allow the machines to learn from data [36]. ML was integrated into the IDS to address the growing number of cyberattacks. Using ML, the IDS can learn from heuristic data, adapt over time,

reduce false positives, detect attack patterns, and match new threats. AI/ML-based intrusion detection is essential in an effective and potent cybersecurity, particularly in detecting zero-day attacks. Many ML techniques are used in IDS. However, there is no clear consensus on which classifiers perform best. Studies focused on the integration and interpretability of the model have offered different results for AI/ML. No AI/ML model stands out as the top one and they differ in accuracy, stability, and transparency [39–42].

The following are some of the fundamental ML algorithm and techniques that can be used in IDS:

4.1.1 Logistic regression

Logistic Regression (LR) is a popular ML algorithm for binary classification tasks found in IDS. It estimate the probability of an event as a number between 0 and 1 using a sigmoid logistic function. Its simplicity, interpretability, and efficiency has made it a common choice in binary classification tasks [43–46]. Sigmoid logistic function (Eq. 1):

$$\boldsymbol{p}(\boldsymbol{x}) = \frac{1}{1 + \boldsymbol{e}^{-(\boldsymbol{w} \cdot \boldsymbol{X} + \boldsymbol{b})}} \tag{1}$$

$\boldsymbol{p}(\boldsymbol{x})$ The probability of the class is positive.($\boldsymbol{w}$) indicates the weight (coefficients) of each feature.($\boldsymbol{X}$) is the feature vector.($\mathbf{b}$) is bias.

4.1.2 KNN (K-nearest neighbors)

The KNN algorithm is a simple ML technique used in the IDS to classify network behavior into normal or attack. It is based on the principle that similar samples are close to each other in the feature space. Applied to IDS, the KNN analyses the k closest labeled sample when a new network event is observed. These nearest neighbors in the data set usually include known patterns of normal and malicious activity. The major class within this neighbors determines the class of the new event. The distance between objects x and y is usually determined by the Eq. (2) [47]. The key advantages of KNN are simplicity and not requiring training, which enable real-time detection. However, KNN researchers use optimization and hyperparameter tuning to enhance its performance, particularly decreasing false positive [39, 43, 48].

$$d_{xy} = \sqrt{\sum_{i=1}^{n} (X_i - Y_i)^2} \tag{2}$$

4.1.3 Naive bayes

Single class naive Bayes is a probabilistic classifier who uses Bayes theorem. It assumes features are independent to effectively classify network events in the IDS. Its speed and simplicity make it a valuable tool capable of real-time intrusion detection. Despite assuming features independence, the naive Bayes algorithm can effectively manage large datasets and be adapted to continuous data through modification such as Gaussian Naive Bayes. Dimension reduction techniques such as principle component analysis (PCA) are often integrated into this algorithm to enhance detection capabilities. IDS solutions based on naive Bayes models are usually evaluated using standard dataset such

as NSL-KDD to ensure maintaining high accuracy and low false positive in detecting a wide range of cyberthreats [44, 48–51].

4.1.4 Decision trees

Decision trees (DTs) are fundamental components of an IDS who serve both classification and regression purposes. They divide the feature space into separate regions based on features values. Structurally, they resemble a tree where each internal node tests a feature, each branch represent the test result, and each leaf represents a class label or classification decision. The algorithm chooses the best feature to partition data based on the acquired data or other criteria and the process continues recursively until the tree reaches its full growth. A key advantage of DTs is their interpretability. Their flowchart-like structure facilitates easy understanding and clear visualization. DTs are used in the IDS to classify data into 'normal' or 'attack' and for regression tasks [39, 43, 44, 48].

4.1.5 Random forest

The Random Forest (RF) algorithm is a well-known ML method that excels in classification tasks in the IDS. Its effectiveness stems from its capacity to process large datasets with multiple input variables while maintaining resilience to overfitting. RFs build multiple DTs in the training phase. Each DT is based on a random data subset and independent sampling of input variables. This creates a diverse forest. During classifying new network activities, each DT votes on the sample classification. The result is determined by majority vote. This algorithm excels both abuse detection, which involves detecting established attack patterns, and anomaly detection, which focuses on deviations from normal behavior. The high accuracy and ability to automatically detect new intrusion patterns without requiring manual coding of the rules, makes RF particularly valuable [44, 46, 48, 49, 52, 53].

4.1.6 SVM

As a potent ML tool, the support vector machine (SVM) improves the capabilities of classification and high-accuracy prediction in complex datasets. In an IDS, the effectiveness of SVM begins with feature selection, where attributes relevant to intrusion activities are chosen, and unnecessary ones are discarded to simplify the model. SVM learn how to detect normal network activities from malicious ones from a labeled dataset. It tries to find an optimal hyperplane that maximizes class separation. The construct excels in specifying data point in stark contrast with the established normal patterns which it labels as potential threats. It is also very apt in abuse detection and classifies new network activities based on learned attack patterns [39, 43, 44, 46, 48–53].

4.1.7 K-means

The K-means algorithm is a core element in clustering techniques, which is aptly applied to the IDS to differentiate between normal and malicious network traffic. Data preparation begins with converting string values to numeric ones and normalizing the data to ensure uniformity across the features. During clustering, the algorithm partitions the dataset into 'k' clusters by assigning data points to the nearest center and calibrating the centers until they become stable. This benefits anomaly detection in the IDS as outliers to these clusters might indicate potential intrusion. K-means is used in the IDS along

with innovative approaches to increase its efficiency. Methods combining metaheuristic algorithms such as atom search organization (ASO) and equilibrium optimization (EQ) enhance feature selection and accuracy and decrease error rate [39, 48, 52, 54].

4.1.8 Ensemble learning

As a central strategy in the IDS, Ensemble learning (EL) combines several ML models to enhance overall performance and improve diagnostics capabilities. The combined power of various algorithms outperforms each on of them which often translates to better accuracy and robustness [39, 55, 56]. EL algorithms in IDS fall into several categories:

Bagging: Like DT, this technique creates multiple models and trains each of them on a random subset of training data. Various subsets of input data are selected using random sampling with replacement, meaning that a data point can appear in multiple subsets. It trains several base ML learners. The final prediction combines the predictions of all models, usually by majority vote, to create a powerful classifier. Bagging reduces variance, prevents overfitting, and enhances the model [57, 58].

Boosting: Boosting is a powerful ML technique that turns weak learners into strong ones. It builds successive models where each new model tries to correct previous errors. It prioritizes misclassified cases which focuses the next models on hard cases. However, overfitting makes it less suitable for noisy data. AdaBoost, Gradient Boosting, and XGBoost are a few of existing boosting algorithms. Boosting has proven to reduce bias and variance which improves accuracy [57–59].

Stacking: is training several different models and using another model, called the meta-model, to combine their predictions. The base models are trained first and their predictions becomes the input to the meta-model. This approach uses the advantages of each base model which potentially enhances the performance compared to the performance of each base model [58–60].

4.2 Deep learning

DL is a subset of ML which uses multilayer NNs to learn from data. This is particularly helpful in learning unstructured data such ad images and text [11, 61]. DL has improved IDS solutions substantially and offers complex capability to detect cyberthreats. It particularly excels in processing big data, detecting complex patterns and anomalies that indicate security breaches [62, 63]. The automatic learning capability and determining the most relevant features for classification tasks make it invaluable in detecting zero-day misuses and advanced persistent threats [64]. Integrating DL into conventional IDS techniques creates a more comprehensive defense mechanism against the landscape of evolving cyberthreats and ensures digital infrastructures remain protected. A DL model is mainly used to reach the least error in the fastest time possible. Figure 4 shows the typical composition of a DL model [65].

Here are some useful and effective DL algorithms integrated into the IDS:

4.2.1 Multilayer perceptrons

Multilayer perceptron (MLP) is a densely interconnected NN to achieve strong fitting capabilities in nonlinear systems. These are used in IDS to classify network traffic and detect anomalies. It consists of an input layer, one or more hidden layers, and an output layer [47, 66], Fig. 5. Each layer is composed of neurons that use activation functions

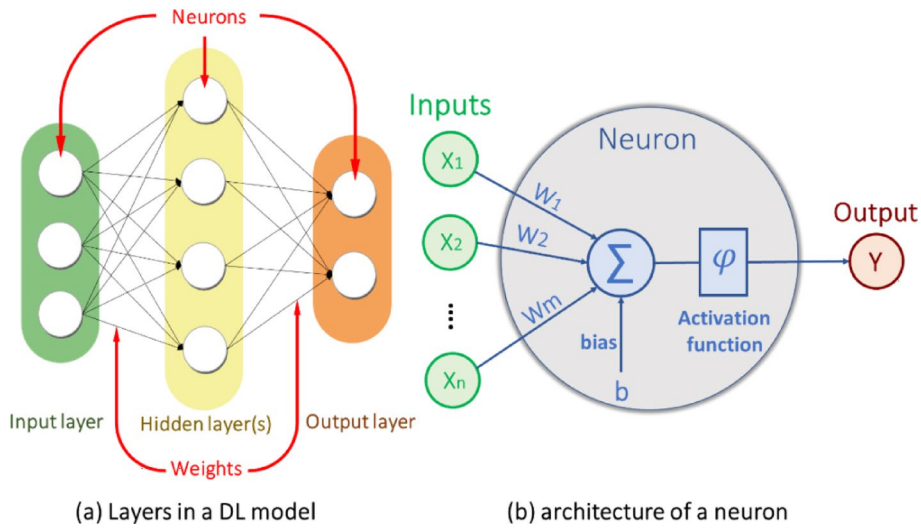

Fig. 4 **a** The general structure of a deep learning model and **b** the structure of a neuron

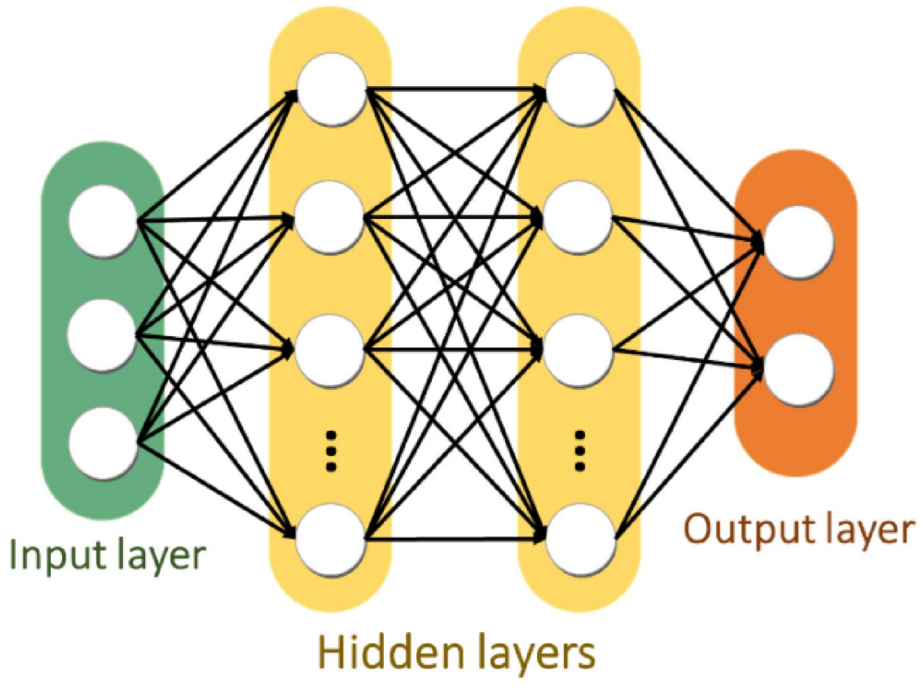

Fig. 5 Structure of the MLP algorithm

to process signals. Post-propagation supervised learning is used to train the NN. MLP maps a set of input data.

to a proper set of outputs. The processing is inspired by the information processing in neuro-biological systems [67]. It is trained on network data to learn the characteristics of normal and malicious traffic in the IDS. Once trained, it can classify new data as normal or intrusive. MLPs are known for their ability to handle big and complex datasets which makes them a valuable and proper tool in combating evolving cyberthreats in network security applications.

4.2.2 Convolutional neural networks

Convolutional Neural Networks (CNN), a subset of DL, is effectively employed in IDS solutions. CNNs are integrated into IDS for their strong feature extraction capabilities especially in complex and large-scale data such as network traffic. Their ability to automatically identify important features without manual intervention is crucial for detecting complex patterns in network traffic [39, 68, 69]. The CNN architecture consists of five main layers (Fig. 6) [64, 66, 70]: The input layer receives the normalized array data. The hidden layer consists of a convolution layer (which uses a convolution kernel to extract features), activation functions (the linear mathematical functions used to control the linear or non-linear output), a pooling layer (for feature selection and information filtering), and a fully connected layer (to minimize feature information loss). Once features extracted, a feature map is sent to the pooling layer which performs sampling and effectively prevents overfitting. Finally, the output layer directly classifies each feature [66]. Convolution layers automatically and adaptively learn spatial hierarchy of features from input data. IDS then uses these features to detect patterns indicative of network intrusion. Integrating CNN into the IDS helps detecting subtle patterns in the network traffic which could indicate malicious activities and are missed by conventional ML methods [70–73].

4.2.3 Deep belief network

Deep belief networks (DBNs), a type of NN, are used in IDS solutions. They are good at detecting complex patterns in large-scale data which is typical in network traffic. DBN is based on the building blocks of Restricted Boltzmann Machines (RBM) (Fig. 7). It consists of multiple stacks of RBMs connected sequentially. RBMs are essential to DBN and are widely used in data reconstruction, data classification, and pattern recognition [74, 75]. Each RBM model performs a nonlinear transformation on its input vectors and produces output vectors that serve as input for the next RBM model in the sequence. It is a two-layer model (input and hidden layer) with data flow in both directions [25, 39, 66, 76]. Except for the first and last one, each layer acts as a hidden layer for the previous nodes and as a visible/input layer for the subsequent nodes. In DBN, each node is connected to each of the other nodes in the previous and next layers. However nodes are not connected to the other nodes in the same layer [25]. DBNs are used in an IDS for feature extraction and classification tasks. Their ability to learn complex representation of data plays a crucial role in increasing the accuracy of intrusion.

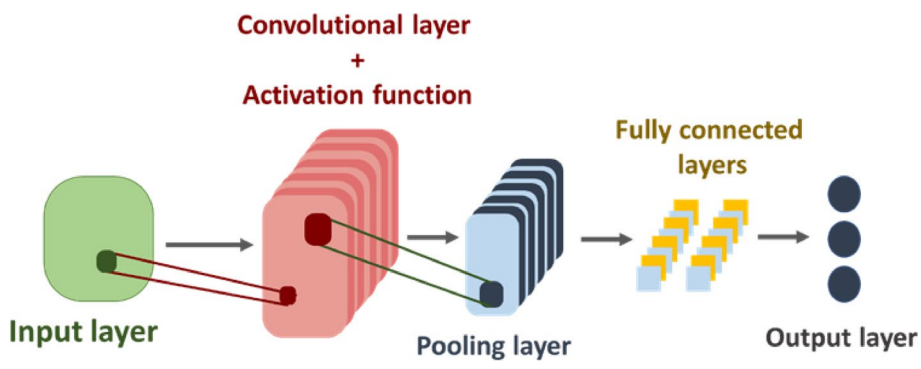

Fig. 6 Structure of the CNN algorithm

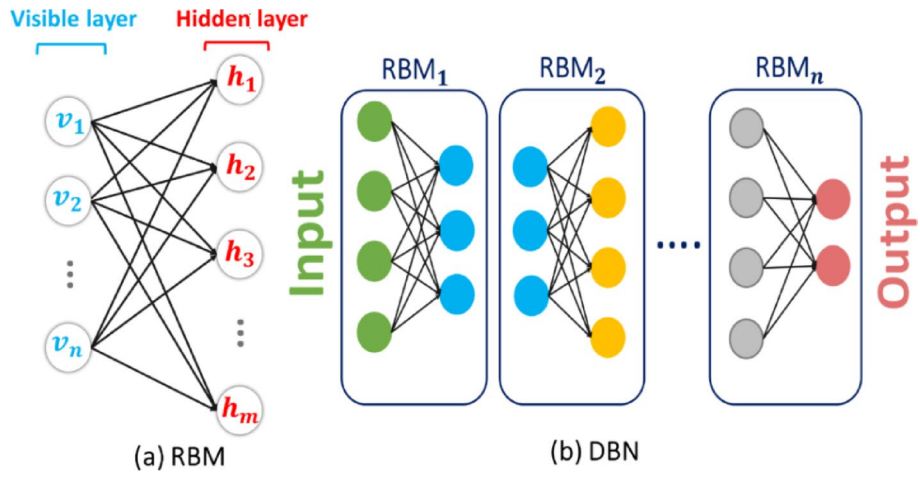

Fig. 7 Structure of Restricted Boltzmann Machine (RBM) and Deep Belief Network (DBN). **a** The architecture of an RBM, showing the visible and hidden layers. **b** The structure of a DBN, composed of stacked RBMs for feature extraction and classification in IDS

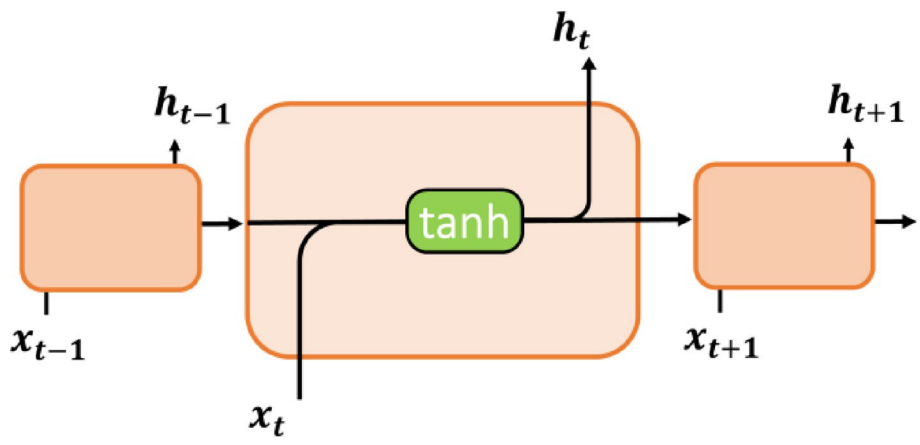

Fig. 8 The structure of a typical RNN

4.2.4 Recurrent neural networks

Recurrent Neural Networks (RNNs) are a specialized type of NN designed to process sequential data. They excel at time series, speech recognition, natural language processing, and translation tasks thanks to their cyclic data flow (propagating information back and forth). Unlike conventional feed-forward NNs, RNNs implicitly keep the previous inputs in their memory which makes them suitable for modeling time dependence. They temporarily keep information for later use by looping through layers [2, 74, 77–79]. However, their limited handling of long sequences can lead to information disappearance (also known as gradient disappearance) [66, 75, 77, 80]. Introducing LSTM and GRU into RNN fixes that issue [2]. RNNs are best suited to IDS solutions that handle sequential data, such as network traffic or time series data. Their unique ability in processing input sequences while having some kind of memory through their internal state allows them to predict not only based on the current input, but also based on already processed data [66, 77]. Figure 8 depicts the structure of a typical RNN. RNNs can be trained to learn normal and malicious pattern of network behavior in an IDS. They detect anomalies by analyzing the sequence of network packets and detecting deviations

from a defined normal pattern [77]. This is particularity useful in detecting complex attacks that unfold over time such as advanced persistent threats (APT) which might be missed by other algorithms that do not consider the temporal sequence of events. RNNs face information (gradient) disappearance when handling long sequences [11, 66, 75, 77, 80, 81].

4.2.5 Long short-term memory (LSTM)

Long Short-Term Memory is a type of recurrent neural network (RNN) designed to address the vanishing gradient problem found in traditional RNNs. Unlike classical RNNs that face the challenge of long-term dependencies, LSTMs can process data sequentially, remember long-term dependencies, and maintain hidden states across thousands of time steps [1, 2, 11, 75, 77, 81–84]. They are applied to tasks such as handwriting recognition and machine translation. LSTM essentially excels in detecting long-term dependencies which makes them an ideal choice for sequence prediction [82, 83]. The LSTM architecture makes it a good choice for solving the RNN gradient problem. It uses a memory cell that represents long-term dependencies in sequential data. LSTM cells can remember the past sequence of data and combine the information with the current input sequence [1]. Each cell consists of three gates, namely the input gate, the output gate, and the forget gate [1, 66, 77, 81–84], Fig. 9:

The forget gate examines the input data and the data received from the previous hidden layer, then uses the sigmoid function to decide which information to remove from the cell (1 means keep, 0 means remove) [11, 66, 77, 81, 84].

$$f_t = \sigma\left(W_f.\left[h_{t-1}, x_t\right] + b_f\right) \tag{3}$$

f_t : Forget gate vector.W_f : *Weight matrix for forget gate.*b_f : Bias for forget gate.x_t : *input at time step t.*h_{t-1} : *previous hidden state.*

The input gate first decides which information should be added and updated using a sigmoid function. Then, a Tanh layer proposes a new vector to be added to the LSTM cell. The cell state is then updated using the new vector values [11, 66, 77, 81, 84].

$$i_t = \sigma\left(W_i.\left[h_{t-1}, x_t\right] + b_i\right) \tag{4}$$

$$\tilde{C}_t = Tanh\left(W_c.\left[h_{t-1}, x_t\right] + b_c\right) \tag{5}$$

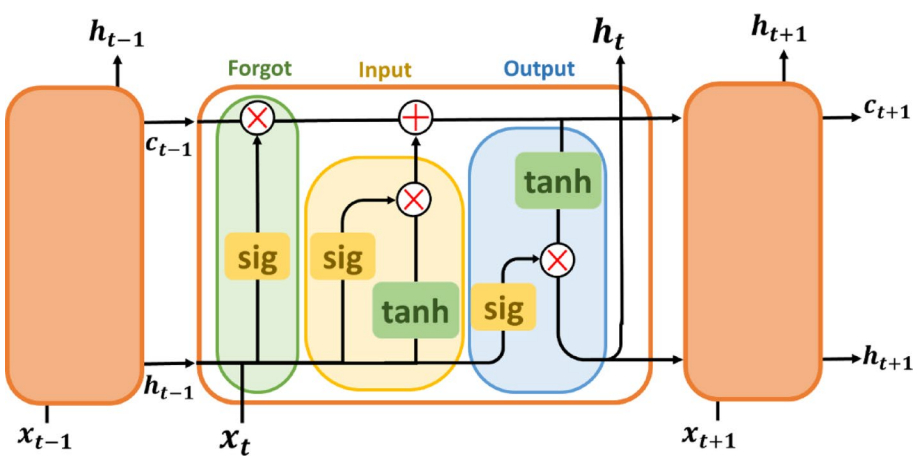

Fig. 9 The structure of LSTM and its gates

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t \tag{6}$$

i_t : Input gate vector.W_i : Weight matrix for input gate.b_i : Bias for input gate.$\widetilde{C}_t$: Candidate cell values.W_c : Weight matrix for candidate state.b_c : Bias for candidate state.C_t : Updated cell state.C_{t-1} : Previous cell state.

The output gate controls what information is transferred from the memory cell to the output by executing a sigmoid function. The result is passed through a Tanh layer so that only the information intended for transmission to the next neuron are sent to the output [11, 66, 77, 81, 84].

$$O_t = \sigma\left(W_o \cdot [h_{t-1}, x_t] + b_o\right) \tag{7}$$

$$h_t = O_t \times Tsnh\left(C_t\right) \tag{8}$$

O_t : Output gate vector.W_o : Weight matrix for output gate.b_o : Bias for output gate.

LSTMs excel at recognizing patterns over time, which is valuable for detecting persistent threats in network traffic. Their ability to learn from data sequences makes them ideal for detecting anomalies in network behavior. LSTM is used in the IDS for analyzing network events to detect unusual patterns. They are particularly effective in environments where attacks happen over a long time and allow the system to detect complex threats and respond to them. An Improved feature selection which focuses on relevant features increase their accuracy and allows LSTM to focus on the most relevant features.

4.2.6 Auto encoders (AE)

The autoencoder neural network is used in unsupervised learning. Its architecture (Fig. 10) consists of an encoder that compresses the input data into a lower-dimensional representation (a bottleneck) and a decoder that reconstructs the data from its encoded form into the original input. During training, the autoencoder minimizes the difference between the input and the reconstructed output [35, 39, 51, 66, 71, 85]. The goal is to

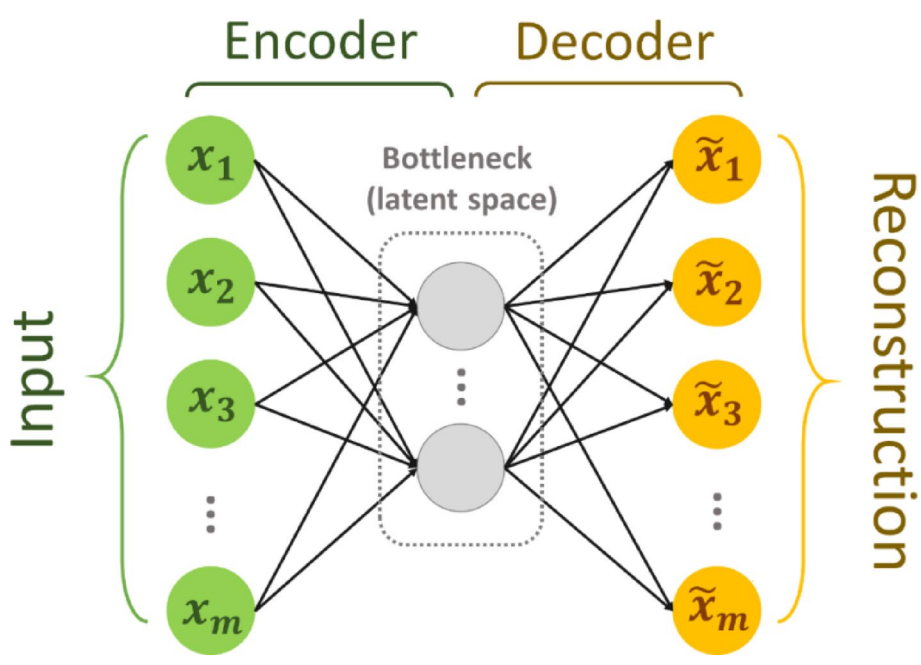

Fig. 10 The architecture of AE

reconstruct the input data at the output with the lowest possible reconstruction error between the input and output data [86, 87]. AEs are especially popular in anomaly detection as they increase accuracy significantly [88]. They compress network traffic data into a lower-dimensional space and highlight anomalies. This process helps AE learn normal data patterns and then detect deviations from this norm, which may indicate a cyberattack [35]. Ease of training and requiring low computational complexity are other advantages of the AEs. The AE architecture can manage large-scale data and select features [84, 89].

4.2.7 Graph neural networks (GNNs)

Graph Neural Networks (GNNs) are deep learning models designed for graph-structured data, representing entities as nodes and relationships as edges in domains like intrusion detection systems (IDS), social networks, bioinformatics, and finance. Unlike CNNs and RNNs, which handle Euclidean data like images or sequences, GNNs capture complex non-Euclidean relationships through message-passing, aggregating neighbor information to update node representations. These models support tasks such as node classification, link prediction, and graph classification, enabling applications like IDS (where devices or packets are nodes and communications are edges), personalized recommendations (e.g., Alibaba, Pinterest), and drug discovery [90, 91].

4.3 Key GNN models and applications

- *Graph convolutional networks (GCNs):* Aggregate neighbor features using normalized graph structures, excelling in node classification (e.g., Cora dataset) and IDS by modeling network traffic patterns (e.g., UNSW-NB15 dataset). Stable for structured data, GCNs struggle with diverse attack detection due to their reliance on similar node connections [92].
- *Graph attention networks (GATs)*: Apply self-attention to prioritize relevant neighbors, ideal for diverse networks. They perform well in node classification (e.g., Cora) and fault diagnosis (e.g., hardware-in-the-loop datasets), offering interpretability despite higher computational demands [92].
- *GraphSAGE:* Uses neighbor sampling for scalable, adaptive learning, adept at IoT attack detection (e.g., ToN_IoT dataset). Efficient for large-scale IDS, it may miss broader graph patterns [92].
- *Advanced models:* Techniques like GraphSMOTE, NRGNN, and DP-GNN enhance GNN robustness, addressing real-world issues such as imbalanced data, noise, and privacy in applications like fraud detection, protein interaction analysis, and secure network modeling [91, 93].

4.4 Adversarial machine learning

Adversarial Machine Learning (AML) involves techniques where attackers exploit vulnerabilities in machine learning models to manipulate their behavior, compromise performance, or extract sensitive information. AML attacks, such as evasion and poisoning, pose significant risks to AI.

systems in critical applications like healthcare and autonomous vehicles [94–96].

4.5 Evasion attacks

Evasion attacks manipulate input data during model deployment to cause misclassifications, e.g., altering traffic signs to mislead autonomous vehicles [96, 97]. Key countermeasures include:

- *Adversarial training*: Enhances robustness by training with adversarial examples. Effective and scalable but computationally intensive [98, 99].
- *Formal verification:* Uses mathematical proofs to ensure model reliability. Offers strong guarantees but is less scalable for large networks [100, 101].
- Less effective methods like randomized smoothing should complement these [102].

4.6 Poisoning attacks

Poisoning attacks corrupt training data to degrade performance or embed backdoors, as seen in the Tay chatbot incident [103]. Key countermeasures:

- *Training data sanitization*: Removes malicious data before training. Proactive, scalable, and widely applicable [104, 105].
- *Adversarial training*: Bolsters resilience but is secondary to sanitization.
- Other methods (e.g., differential privacy, model pruning) are less effective standalone due to scalability or accuracy trade-offs [106, 107].

4.7 Relevance

These defenses, aligned with NIST [94] and MITRE ATLAS [95], are critical for securing ML models against real-world threats. Tools like Adversarial Robustness Toolbox [108] aid implementation, though scalability and accuracy trade-offs remain challenges.

4.8 Federated learning

Federated Learning (FL) enables collaborative training of AI models across multiple clients while keeping data localized, enhancing privacy in applications like healthcare and autonomous systems [109].

Federated averaging (FedAvg): FedAvg aggregates local model updates from clients, computing a weighted average to update a global model. This reduces communication costs but struggles with data heterogeneity, which can bias models. Variants like FedProx mitigate this by adjusting for client data differences.

Differential privacy (DP): DP adds noise to model updates to prevent data inference, aligning with XAI's transparency goals [42]. It ensures privacy with a quantifiable budget, but excessive noise may reduce model accuracy, impacting explainability in complex systems.

Homomorphic encryption (HE): HE allows computations on encrypted data, ensuring secure aggregation. It supports XAI by protecting sensitive data during model explanation [109], but its computational cost limits scalability for large models.

Challenges: Federated Learning (FL) struggles with data heterogeneity, causing biased models, and privacy-accuracy trade-offs from noisy updates, hindering explainability. High computational costs of encryption limit scalability. Integrating FL with XAI demands efficient, transparent explanations for safety–critical systems [109].

4.9 Explainable AI techniques

Explainable AI (XAI) techniques enhance transparency and trust in AI models by elucidating their decision-making processes, critical for safety–critical domains like healthcare and autonomous vehicles [2]. By addressing the black-box nature of complex models, XAI mitigates biases, ensures accountability, and supports regulatory compliance, enabling users to understand and validate predictions [110, 111].

SHAP (Shapley additive explanations): SHAP is a model-agnostic method that quantifies feature contributions to predictions using Shapley values from cooperative game theory. It calculates the average impact of each feature across all possible feature subsets, providing consistent and precise explanations. For example, in healthcare, SHAP can reveal which patient features drive a diagnosis prediction. However, its computational complexity grows exponentially with feature count, limiting scalability for high-dimensional models [111, 112].

LIME (*Local interpretable model-agnostic explanations*): LIME [113] generates local explanations by approximating complex models with simpler, interpretable surrogates, such as linear models. By perturbing input data and observing output changes, LIME identifies influential features for individual predictions. For instance, in text classification, LIME highlights key words affecting sentiment predictions. Its simplicity aids interpretability, but performance can degrade with high-dimensional or highly non-linear data, and results may vary with perturbation strategies.

Attention mechanisms: Attention mechanisms in Transformer models assign weights to input segments, highlighting their relevance to predictions [111]. For example, in natural language processing, attention weights reveal which words drive a model's output, visualized as heatmaps. While this enhances local interpretability, the complexity of multi-head attention can obscure global model behavior, necessitating advanced visualization or propagation techniques for clear explanations in safety–critical applications.

Challenges: SHAP and LIME face scalability challenges due to computational demands, particularly for large datasets or complex models. Attention mechanisms require refined methods to distill complex interactions into clear explanations, especially in safety–critical contexts where transparency is paramount. Balancing interpretability with accuracy and developing efficient algorithms remain critical for deploying these techniques in real-time, high-stakes applications. [111].

4.10 Choosing the best model for the IDS

Previous studies in IDS has shown that certain models have a strong performance in anomaly detection. This models are effectively helpful in detecting unusual behaviors that could indicate cyberattacks. However, there is no known best AI/ML/DL model. The effectiveness of each model depends on various factors, including the specific context and the type of attacks they face. Different models may excel in different scenarios. Tests have shown that a combining strategy can improve prediction accuracy. Combining several classifier outputs enjoys their complementary strengths and improves the overall performance. However, such merging needs to be performed carefully as the addition of weak classifiers could be counterproductive, potentially create noise and false predictions, and reduce the accuracy of the final combined result.

The selection of ML and DL models for IDS depends on several factors, including the type of attack, dataset characteristics, and performance requirements. For instance,

signature-based attacks are best detected using models like Random Forest or SVM, which excel in classifying known patterns with high precision [44]. In contrast, anomaly-based detection, particularly for zero-day attacks, benefits from DL models like LSTM or Autoencoders, which can learn complex temporal and spatial patterns in network traffic [77]. The choice is also influenced by computational resources, as DL models require significant processing power, and the trade-off between false positives and detection accuracy. For example, in environments with high normal traffic, models with low false positive rates, such as Decision Trees, are preferred to minimize disruptions.

5 Evaluation criteria

Precision, f-measure, recall, accuracy, false positive, and false negative are criteria that are generally used to evaluate IDS models and compare their performances. These metrics are crucial for understanding the effectiveness of a classifier in particular in cybersecurity where distinguishing between normal behavior and attacks is of eminent significance. These values are calculated using the confusion matrix (Fig. 11) [1, 2, 21, 45, 63, 77, 82, 114, 115]:

- *True positive (TP)*: Abnormal records correctly classified as anomaly (A hacker tries to break into the server, and the IDS correctly raises an alert).
- *True negative (TN)*: Cases correctly classified as normal behavior (An employee checks their email, and the IDS does not raise any alert).
- *False negative (FN):* Actual abnormal cases considered normal behavior by the classifier (e.g. mistaking an attack for a normal behavior, e.g. A virus spreads in the network, but the IDS does not notice it). False negatives pose a serious security risk, allowing attackers to exploit vulnerabilities without detection, potentially resulting in data breaches, system compromise, and prolonged unauthorized access.

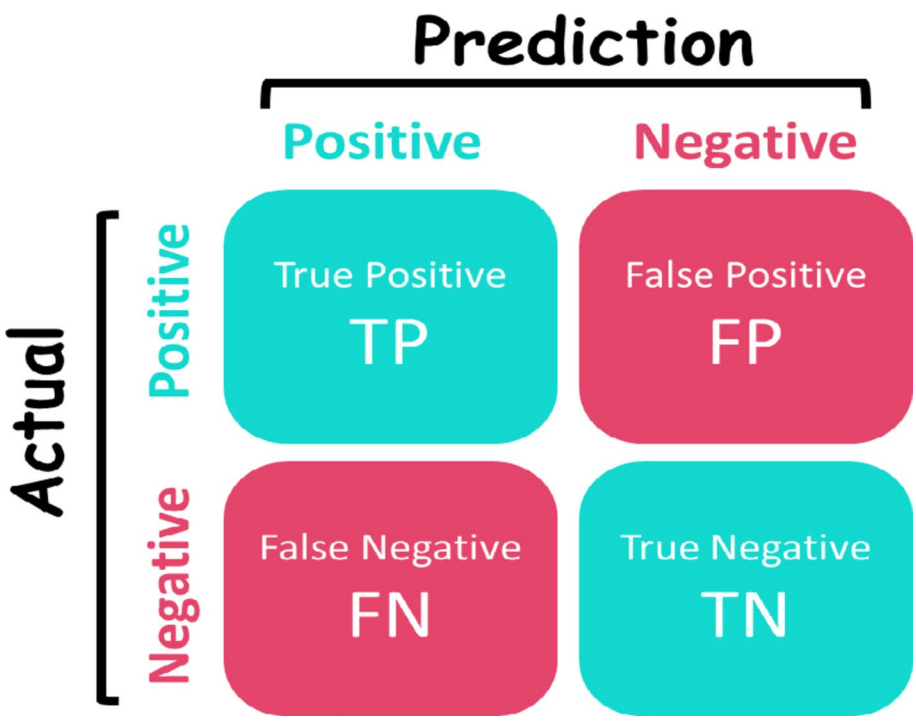

Fig. 11 Confusion matrix

- *False positive (FP):* Actual normal behaviors detected as anomaly by the classifier (normal behavior mistaken for anomaly, e.g. A user downloads a company file, and the IDS wrongly thinks it's a data theft attempt). High false positive rates can overwhelm security teams with unnecessary alerts, leading to alert fatigue, wasted resources, and potential neglect of genuine threats.
- *Accuracy:* The ratio of correctly detected records to the entire test dataset A higher accuracy means better ML model. However, this might reflect the true performance of the model in an unbalanced dataset.

$$Accuracy = \frac{TP + TN}{(TP + TN + FP + FN)} \tag{9}$$

- *Precision:* The ratio of samples correctly classified as attacks to all samples classified as attacks. This measurement focuses on reducing false positive and is useful when false positives have a high impact of the model's quality, e.g. In spam detection.

$$Precision = \mathrm{TP}/\left(\mathrm{TP} + \mathrm{FP}\right) \tag{10}$$

- *Recall (or sensitivity):* The ratio of samples correctly classified as attacks to all samples predicted to be attacks. High recall is useful when false negatives has a high impact on the model's quality, e.g. In detecting fraud and medical diagnosis.

$$Recall = \mathrm{TP}/\left(\mathrm{TP} + \mathrm{FN}\right) \tag{11}$$

- *F1-Score (F1-Measure):* This criteria uses harmonic mean instead of the arithmetic mean. It is the harmonic mean of the precision and recall and is more effective in determining the performance on unbalanced datasets. A higher F1 score means better ML model.

$$F1 - Score = \frac{(2^{*}\mathrm{Precision}^{*}\mathrm{Recall})}{(\mathrm{Precision} + \mathrm{Recall})} \tag{12}$$

6 Datasets

When it comes to IDS, datasets play an essential role in training and evaluating the intrusion detection models and help researchers enhance the security of the systems [6, 19]. An intrusion detection dataset is a collection of information from various sources including network traffic streams. These streams contain details about the host, user behavior, and system settings. The goal is to analyze attack patterns and detect unusual network activity. Network flow analysis, involving examining packet information such as IP addresses, port numbers, and service types, is performed on the collected traffic. System configurations and user data, which are not accessible through flow analysis, are obtained from the network host. For example, failed login attempts can indicate intrusion activity [16].

6.1 The evolution of intrusion detection datasets

6.1.1 DARPA 1998 dataset

This is one of the oldest and best-known datasets, created and developed by MIT Lincoln Laboratory in 1998 for IDS evaluation. It is a mix of normal and attack traffic that simulates a military network environment [16].

6.2 KDDCup99

This is a benchmark dataset widely used in the field of IDS. It is a modified version of DARPA 1998. The dataset includes a wide range of intrusions and has been widely used to evaluate IDS models. Despite its popularity, it has been criticized for its redundancy and old attack types [16].

6.3 NSL-KDD dataset

This dataset overcomes the limitations of KDD'99 dataset. Unlike KDD'99 which suffered from duplicate data and bias toward them, It ensure a fair representation. Eliminating duplicate cases and appropriate selection of records allow various ML algorithms achieve better classification rates. Despite their smaller training size, they hold enough data to train ML models [6, 12, 16, 81, 116].

6.4 UNSW-NB15 dataset

Moustafa et al. from the Cyber-Range Lab in Central Australia created the UNSW-NB15 dataset using the IXIA PerfectStorm tool in 2015 [84, 117, 118]. Unlike older datasets such as NSL-KDD/KDDCup-99, UNSW-NB15, it includes modern attack patterns and standard network traffic. The dataset contains 49 features (excluding class labels) and a total of 2,540,044 records, including ten class labels, of which nine are attack class labels and one is regular label [114, 118, 119]. Data was collected from three real-world websites: BID (Symantec Corporation), CVE (Common Vulnerabilities and Exposures), and MSD (Microsoft Security Bulletin) and was then simulated in a laboratory environment to create the dataset [13, 120]. Its key advantages over older datasets are large number of attacks and up-to-date information [121].

6.5 CICIDS 2017 dataset

This dataset, developed by the Canadian Institute for Cybersecurity, includes a wide range of attack scenarios and normal traffic, providing a comprehensive benchmark for evaluating IDSs [122]. It is a five-day collection of normal traffic and attack data from the Canadian Cybersecurity Institute. The gathered data is divided into 8 items. Attacks performed during the period is different for each day of the week [85]. This dataset was created to address the lack of real-time network traffic datasets for intrusion detection evaluation. It outperforms other datasets, such as NSL-KDD and UNSW-NB15, with its extensive feature set of over 80 features, making it stand out [81, 121].

6.6 CSE-CIC-IDS2018 dataset

This is a joint effort between the Communications Security Institute (CSE) and the Canadian Cybersecurity Institute (CIC). It serves as a benchmark for IDS and provides a comprehensive set of network traffic data. The dataset consists of 80 features including labels. Most of the features are flow-based statistical traffic information extracted with CICFlowMeter [62]. It includes seven attack scenarios: Brute-force, Heartbleed, Botnet, DoS, DDoS, web attacks, and network intrusion from the inside. The attack network consists of 50 machines, while the victim organization consists of 420 PCs and 30 servers [121].

6.7 TON_IoT dataset

This dataset focuses on Internet of Things (IoT) environments, demonstrating the increasing importance of securing IoT devices and networks. This relatively new and extensive dataset was created in 2019 by Abdullah et al. [123]. It includes different types of IoT data, such as operational reports of the system, telemetry data for IoT/IIoT services, as well as IoT network traffic. The data is collected from a medium-size network in Cyber Range and IoT laboratories at UNSW Canberra (Australia) [124].

6.8 OD-IDS2022 dataset

Patel, Mehtre, and Wankar [124] introduced the OD-IDS2022 dataset, which is designed for classification of ML-based attack in IDS. It satisfies the requirement for a comprehensive and reliable dataset in IDS research by including both types of malignant attacks and another twenty-eight common cases and satisfies the eleven desirable characteristics of IDS datasets, such as attack diversity, complete traffic, and metadata. The authors compare the performance of four ML algorithms (random forest, decision tree, naive Bayes, and support vector machine) on the OD-IDS2022 dataset and found that the support vector machine (SVM) algorithm had the highest prediction accuracy [125]. It stands out among the other datasets by including the latest attacks and offering a more comprehensive presentation of network traffic which turns it into a comprehensive option for general IDS research.

6.9 CICIoT2023 dataset

The CICIoT2023 dataset, presented by Neto et al. [125] from the University of New Brunswick, is a comprehensive resource for IoT cybersecurity research, addressing the need for realistic datasets capturing large-scale attacks. Generated using a topology of 105 real IoT devices, it includes 33 attacks categorized into seven classes: DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, and Mirai. These attacks, executed by malicious IoT devices targeting others, cover scenarios like UDP floods, Slowloris, SQL injection, and Mirai botnet variants. Data was collected using Wireshark and a network tap, with features extracted for machine learning (ML) and deep learning evaluations. The dataset supports binary, multi-class (8 and 34 classes), and grouped attack classification, with Random Forest and Deep Neural Networks achieving high accuracy (up to 99.6% for binary classification). CICIoT2023 enables the development of advanced security analytics for IoT environments [126].

6.10 CICIoV2024 dataset

Neto et al. [126] examined the CICIoV2024 dataset, a groundbreaking benchmark designed to advance real-world Intrusion Detection System (IDS) approaches against Denial-of-Service (DoS) and spoofing attacks in the Internet of Vehicles (IoV) environment. Created through extensive tests on the electronic control unit (ECU) of a 2019 Ford automobile, the dataset provides an accurate view of intra-vehicular communications via the Controller Area Network (CAN) protocol. The authors highlight the complexity of network traffic in IoV operations and the need for methods to identify complex patterns, conducting five targeted attacks—DoS and various spoofing types (Steering Wheel, RPM, Gas, Speed)—against the vehicle's fully intact internal structure. With 1,408,228 instances across binary, decimal, and hexadecimal formats, the dataset

captures realistic traffic from benign operations and supports machine learning (ML)-based cybersecurity solutions. ML evaluations show strong performance from Deep Neural Networks and Random Forest models, despite challenges with class imbalances. It sets a new baseline for IoV security research, ideal for anomaly detection and smart city integration [127].

6.11 CICIoMT 2024

To investigate the security of Internet of Medical Things (IoMT) devices, the CICIoMT2024 dataset provides a robust benchmark for evaluating cyberattack detection and mitigation strategies. Developed by Dadkhah et al. (2024), this dataset captures network traffic from 40 IoMT devices (25 real and 15 simulated) across Wi-Fi, MQTT, and Bluetooth protocols, subjected to 18 diverse cyberattacks, including DDoS, DoS, Recon, MQTT-specific attacks, and spoofing. The dataset's comprehensive profiling of device lifecycles, from network entry to exit, enables anomaly detection tailored to healthcare environments [128].

6.11.1 Dataset usage and preprocessing

The datasets discussed in this survey, including KDD Cup 1999, NSL-KDD, UNSW-NB15, CICIDS2017, CSE-CIC-IDS2018, TON_ IoT, OD-IDS2022, and CICIoV2024, (Table 4) are typically used in their preprocessed forms as provided by their respective creators. Data preprocessing plays a crucial role in Intrusion Detection Systems (IDS), especially when using machine learning or deep learning models. Raw data from network or system sources often requires transformation into a clean and structured format. This process includes cleansing, integration, transformation, and reduction, which remove irrelevant or redundant information and ensure data consistency (Table 5) [129]. While specific preprocessing steps may vary by dataset and study, they are essential for improving model performance and reducing false positives [129]. To illustrate the impact of standard datasets such as CICIDS2017, UNSW-NB15, and KDDCup99 on the performance of intrusion detection systems, Table 6 summarizes the performance of selected intrusion detection models based on machine learning and deep learning techniques. These models are evaluated based on key metrics such as accuracy, detection rate, false positive rate, and F1-score.

While older datasets like DARPA 1998 and KDD Cup 1999, offer historical value and comparability, they often lack modern threats and realistic traffic. Newer datasets such as TON_IoT and CICIoV2024, address these gaps but may still face issues like class imbalance or synthetic traffic.

6.11.2 Ethical considerations

The datasets used in intrusion detection research, such as those discussed in this survey, are designed to comply with ethical standards, including privacy protection and data anonymization. Datasets like CICIDS2017, UNSW-NB15, and others are typically anonymized to remove personally identifiable information, ensuring compliance with privacy regulations. Researchers are encouraged to adhere to ethical guidelines, such as obtaining necessary permissions for dataset usage and ensuring that the data is used solely for research purposes, to maintain trust and integrity in cybersecurity studies.

Table 4 Comparative summary of benchmark datasets for intrusion detection systems

Dataset	Year	Source	Features	Records	Attack types	Purpose / highlights
DARPA 1998	1998	[130]	Simulated TCP/IP traffic	~4.9 million (approx.)	DoS, probing, R2L, U2R	First comprehensive IDS dataset; designed for military-like network traffic. Generated using simulated attacks in a testbed. Basis for future datasets like KDD99
KDD Cup 1999	1999	[16]	41 features	4,898,431 (train + test)	22 attacks in 4 categories (DoS, R2L, U2R, probe)	Popular for benchmarking ML in IDS. Criticized for class imbalance, outdated attack types, and redundancy. Derived from DARPA but in tabular format
NSL-KDD	2009	[45]	41 features	Train: 125,973Test: 22,544	Same as KDD (without duplicates)	Improved version of KDD99. Removes duplicate records and balances the dataset to better evaluate generalization of ML models. Still based on older attacks
UNSW-NB15	2015	[118]	49 features	2,540,044	9 attack types incl. Fuzzers, Shellcode, Worms, Reconnaissance	Modern traffic using real websites (Symantec, CVE, MSD). Simulated with IXIA tools. Designed to reflect current threats in diverse environments
CICIDS 2017	2017	[122]	80 + features	~2.8 million (approx.)	PortScan, DoS, DDoS, brute-force, infiltration, botnet, etc	5 days of labeled traffic with different attacks per day. High-quality features using CIC-FlowMeter. Reflects enterprise traffic and user behaviors
CSE-CIC-IDS2018	2018	[122]	80 + flow features	16,233,002	7 major attack types (Botnet, DoS, Heartbleed, Web, etc.)	Combines different protocols and attacks in a large-scale network. 50 attacker devices, 420 victim PCs, 30 servers. Ground-truth labels and timing available

Table 4 (continued)

Dataset	Year	Source	Features	Records	Attack types	Purpose / highlights
TON_IoT	2019	[123]	Mixed (network, telemetry, log)	Over 1M	IoT-specific threats: DDoS, MITM, scanning, malware	Captures telemetry from IoT sensors, logs from services, and network traffic. Suitable for federated learning and fog/edge environments
OD-IDS2022	2023	[125]	80 + features	~ 1.6 million (approx.)	28 different attack classes + benign traffic	Focused on comprehensive ML-based IDS testing. Includes metadata and satisfies key criteria: up-to-date, diverse, labeled, and realistic. SVM performed best in benchmark
CICIoT2023	2023	[126]	47 features	46,686,580	includes 33 attacks: DDoS(12 types), DoS (4 types), Recon (5 types), Web-based (5 types), Brute Force, Spoofing, Mirai	33 attacks on 105 IoT devices, supports binary/8-class/34-class classification, up to 99.6% accuracy with ML/DL
CICIoV2024	2024	[127]	CAN-bus data + metadata	1,408,228	DoS, Spoofing (Steering Wheel, RPM, Gas, Speed)	Real vehicle data from Ford 2019 using CAN protocol. Designed for IoV/automotive intrusion detection research. Focuses on high-complexity, time-sensitive vehicle communication
CICIoMT2024	2024	[128]	44 features	7,739,593	18 attacks: DDoS, DoS, Recon, MQTT, Spoofing	IoMT security benchmark, 40 devices (25 real, 15 simulated), ML with up to 99.8% accuracy, multi-protocol traffic, supports healthcare anomaly detection

6.12 Characteristics of a good dataset

A well-built IDS dataset should satisfy several key requirements. First, it must cover diverse attacks of different types to ensure the detection of various intrusions, including known attacks and unknown (zero-day) ones. Next, it should provide a comprehensive representation of network traffic, encompassing both normal and attack patterns. Third, it should give access to the metadata (such as timestamps and IP addresses) that enhances context understanding. Fourth, data accuracy and consistency are crucial for reliable results. Fifth, It should ensure the privacy of sensitive data by de-identification of sensitive data according to privacy protection regulations. Finally, It should be readily available to the researchers, include clear documentation, and require minimal

Table 5 Data preprocessing techniques for intrusion detection systems (IDS)

Preprocessing step	Purpose	Techniques / methods used
Data cleansing	Improve data quality to reduce false positives/negatives	Handling missing values (mean, interpolation, KNN) Removing noisy/inconsistent entries Outlier detection (DBOR) Clustering techniques (e.g., SOHAC, K-means) Regression models, SimpleImputer
Data integration	Combine logs or traffic data from multiple sources (e.g., host-based, network-based sensors)	Schema alignment across sources Redundant record elimination Harmonizing feature formats (e.g., IP formats, timestamps)
Data transformation	Convert data into suitable format/structure for modeling	Normalization (Min-Max, Z-score) Aggregation & Averaging (e.g., Moving Average) Symbolic Aggregate approXimation (SAX) PCA Correlation-based transformation
Data reduction	Reduce data volume and dimensionality while preserving relevant features for intrusion detection	Removing outliers/spikes Log transformation Normalization ([0,1 Feature selection (Mutual Information, AutoCorrelation, CFS, RReliefF) PCA Clustering (e.g., Improved Fuzzy Partition Matrix—IFPM) Wavelet Transform (WT) + Moving Average

preprocessing to make test repeatable. Associating the dataset with a specific case or domain ensures optimal IDS performance.

7 Limitations and challenges of ML and DL based intrusion detection systems

Despite significant advancements in ML and DL for IDS, several key limitations hinder their optimal performance in practical applications. This section examines these challenges.

7.1 High false positive rate

High false positive rates, where normal activities are mistakenly flagged as malicious, remain a common issue in anomaly-based IDS (A-IDS) [22, 23]. This leads to analyst fatigue and resource waste, potentially undermining trust in the system in high-traffic operational environments. Ensemble methods like Random Forest can mitigate this issue, though balancing sensitivity and precision remains a challenge [16].

7.2 High computational requirements and training time

Complex DL models, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), require substantial computational resources and lengthy training times [24]. This restricts their deployment in resource-constrained devices, such as IoT sensors, limiting scalability. Lightweight models like Autoencoders offer a viable alternative, but further research is needed to maintain accuracy [6, 39].

7.3 Difficulty in real-time deployment

The computational complexity of advanced ML/DL models impedes real-time detection, which is critical for timely threat response [6]. Such delays can disrupt performance in high-throughput environments like enterprise networks. Stream processing or edge computing can alleviate this, but further optimization is required [23, 24].

Table 6 Summary of performance for selected intrusion detection models based on machine learning and deep learning technique

Study	Technique	Dataset(s)	Accuracy (%)	Detection rate (%)	False positive rate (%)	F1-score (%)
[45]	DNN (5-layer, ReLU, 1024 units, lr = 0.1) , (Binary)	KDDCup 99	93.5	93.5	6.5	93.5
		NSL-KDD	92.0	92.0	8.0	92.0
		UNSW-NB15	70.0	70.0	30.0	70.0
	DNN (5-layer, ReLU, 1024 units, lr = 0.1) , (Multi-class)	KDDCup 99	92.0	80.0 (Avg)	8.0	80.0 (Avg)
		NSL-KDD	90.0	78.0 (Avg)	10.0	78.0 (Avg)
		UNSW-NB15	68.0	65.0 (Avg)	32.0	65.0 (Avg)
[131]	DNN	UNSW-NB15 (Binary)	99.19	99.19	0.81	99.19
		UNSW-NB15 (Multi-class)	97.04	97.04	2.96	97.04
	RF	UNSW-NB15 (Binary)	98.86	98.86	1.14	98.86
		UNSW-NB15 (Multi-class)	91.77	91.77	8.23	91.77
	GBT	UNSW-NB15 (Binary)	97.92	97.92	2.08	97.92
[132]	K-Means + + & SVM	HTTP	99.35	99.6	0.9	99.35
		TCP	96.20	92.9	0.5	93.54
		Wiki	87.50	76.0	1.0	86.36
		Twitter	97.91	96.0	0.18	97.87
		Email	96.30	93.0	0.4	96.39
[133]	LibSVM	KDDCup99	99.81	99.81	0.37	99.72
	J48 Decision tree		99.95	99.95	0.05	99.90
	J48 + Naïve bayes		99.11	99.11	4.29	97.45
	LibSVM + J48 + Naïve bayes		99.62	99.62	1.57	99.05
[134]	HAST-I (CNN)	DARPA1998	94.75	93.12	2.88	95.06
	HAST-I (CNN)	ISCX2012	95.92	94.65	1.54	96.54
	HAST-II (CNN + LSTM)		95.93	94.65	1.34	96.63
[28]	RHS-RBM (Feature selection + deep RBM)	KDD Cup 1999	99.92	99.88	0.03	99.93
[135]	RBF neural network	KDD Cup 1999	89.0	71.4	3.8	80.0
[121]	LSTM with SPIP framework	NSL-KDD	81.7	70.6	3.0	81.5
		UNSW-NB15	86.6	98.8	3.0	89.1
		ToN_IoT	87.3	88.0	2.5	82.9

Table 6 (continued)

Study	Technique	Dataset(s)	Accuracy (%)	Detection rate (%)	False positive rate (%)	F1-score (%)
[136]	Decision tree	CIC-IDS2017 (Train)	99.59	99.60	0.41	99.59
		CSE-CIC-IDS2018 (Test)	59.42	59.34	40.66	52.82
		LUFlow 2020 (Train)	99.94	99.95	0.06	99.94
		LUFlow 2021 (Test)	99.90	99.90	0.10	99.90
	Random forest	CIC-IDS2017 (Train)	99.67	99.68	0.33	99.67
		CSE-CIC-IDS2018 (Test)	59.49	59.41	40.59	52.91
		LUFlow 2020 (Train)	99.94	99.94	0.06	99.94
		LUFlow 2021 (Test)	99.91	99.91	0.10	99.91
	Support vector machine	CIC-IDS2017 (Train)	96.00	96.03	3.98	96.00
		CSE-CIC-IDS2018 (Test)	75.59	75.56	24.44	75.02
		LUFlow 2020 (Train)	99.56	99.56	0.45	99.56
		LUFlow 2021 (Test)	99.34	99.35	0.66	99.35
	Artificial neural network	CIC-IDS2017 (Train)	95.49	95.52	4.49	95.49
		CSE-CIC-IDS2018 (Test)	70.00	69.95	30.05	68.68
		LUFlow 2020 (Train)	99.60	99.60	0.40	99.60
		LUFlow 2021 (Test)	99.30	99.30	0.70	99.31
	Deep neural network	CIC-IDS2017 (Train)	97.35	97.37	2.63	97.35
		CSE-CIC-IDS2018 (Test)	65.18	65.12	34.88	62.14
		LUFlow 2020 (Train)	99.82	99.82	0.19	99.82
		LUFlow 2021 (Test)	99.75	99.76	0.25	99.75
[137]	ResNet-18 + Learnable MFCC	IoTID20	99.90	99.90	0.10	99.90
		CICIoT2023	99.38	99.38	0.62	99.38
		NSL-KDD	100.00	100.00	0.00	100.00
	ResNet-18	IoTID20	77.03	77.03	22.97	77.03
		CICIoT2023	68.90	68.90	31.10	68.90
		NSL-KDD	99.70	99.70	0.30	99.70
	Swin transformer	CICIoT2023	72.82	72.82	27.18	72.82
		NSL-KDD	99.71	99.71	0.29	99.71
[138]	Transfer learning (ResNet-50 + DNN, SMOTE) (Binary)	CSE-CIC-IDS2018	92.0	91	~13	89
	Transfer learning (ResNet-50 + DNN, SMOTE) (multiclass)		92.0	90 (Avg)	~8	90 (Avg)
[139]	Strategic selection and forgetting (SSF) with continual learning, clustering, and memory buffer updates	NSL-KDD, UNSW-NB15	90.50 (NSL-KDD), 90.27 (UNSW-NB15)	94.79 (NSL-KDD), 93.89 (UNSW-NB15)	Not specified	91.90 (NSL-KDD), 91.40 (UNSW-NB15)

7.3.1 Case studies: IDS in real-world applications

IDS applications in real-world scenarios enhance their practical relevance. In 5G networks, IDS models using CNNs detect DDoS attacks with 96% accuracy on TON_IoT datasets, ensuring robust network security for high-speed connectivity [115]. In healthcare, federated learning-based IDS (FL-IDS) protect patient data by training models across distributed hospital servers, achieving 95% accuracy on IoT medical datasets

while complying with GDPR [20]. These case studies demonstrate IDS adaptability to diverse domains, though scalability and real-time constraints remain challenges.

7.4 Vulnerability to adversarial attacks

ML and DL models are vulnerable to adversarial attacks, where malicious inputs are crafted to evade detection [8]. This threat can compromise IDS reliability, particularly for anomaly-based systems reliant on learned patterns. Adversarial training can enhance robustness, but its development introduces additional computational costs and complexity.

7.5 Dependence on high-quality and labeled data

Many ML models require accurately labeled datasets for effective training. However, in real-world scenarios, data is often unlabeled, inconsistent, or incomplete, which complicates the training and evaluation processes. This challenge is particularly evident in anomaly detection, where anomalous instances are significantly rare than normal ones, leading to issues such as class imbalance [16, 31].

8 Conclusions

IDS is a fundamental requirement in cybersecurity. Research in this domain is critical due to the ever-increasing complexity and frequency of cyberattack. As cyberthreats evolve, conventional security measures such as firewalls and antivirus software often fail to detect and mitigate these advanced threats. By timely detection of malicious activities, IDS creates an additional defense layer and protects the confidentiality, integrity, and availability of data. The increasing zero-day attacks and growing number of malwares highlights the need for a more powerful and intelligent IDS. Integrating ML and DL learning techniques in IDS research brings promising advances in accurate detection of and timely response to the threats. This ongoing research is essential for developing scalable, efficient, and transparent IDS solutions capable of adapting to the ever-changing cyberthreat landscape.

This study provides a comprehensive overview of Intrusion Detection Systems (IDS), emphasizing the distinction between signature-based and anomaly-based IDS, their architectures, and the application of Machine Learning (ML) and Deep Learning (DL) algorithms. Through an analysis of benchmark datasets such as CICIDS2017, UNSW-NB15, and TON_IoT, and a structured comparison of recent studies (Table 2), this survey contributes to a better understanding of current challenges and future directions in IDS research. Key contributions include consolidating existing knowledge, categorizing ML and DL techniques, and identifying research gaps for developing scalable and robust IDS solutions. Future research can focus on improving model transparency and addressing emerging threats.

9 Future directions

IDS domain is evolving rapidly using DL and ML models. However, there are several unexplored areas awaiting future research. Future research should focus on strengthening the robustness of the DL and ML models against adversarial attacks to detect and mitigate intrusion and improve the reliability of IDS in real-world scenarios and achieving minimum latency real-time detection without compromising detection accuracy in

real-time detection. In addition to the challenges outlined, future IDS research must address emerging threats such as AI-based attacks and quantum computing threats. AI-based attacks, including adversarial machine learning, can manipulate ML/DL models by introducing subtle perturbations to input data, leading to misclassification or evasion of detection [140]. Quantum computing poses another significant threat, as quantum algorithms could potentially break current cryptographic protocols, necessitating quantum-resistant IDS solutions [141]. Future research should focus on developing robust defenses against adversarial AI, such as adversarial training and model hardening, and exploring quantum-safe cryptographic methods to ensure IDS resilience in a post-quantum era. As the network continues to grow and expand, scalability becomes an essential concern. The research should aim to develop scalable models of DL and ML that manage large size data effectively. Integrating IDS into other cybersecurity tools and systems could create a more comprehensive defense mechanism. The black-box nature of DL models poses a significant challenge. Incoming research should focus on the explainability and transparency of these models to gain better insights on their decision-making process. Network traffic data is often unbalanced, with normal traffic being significantly more prevalent than malicious activity. Advanced techniques for managing unbalanced dataset deserve more attention. It is necessary to develop models that can generalize across various network environments and attacks. Future studies need to highlight techniques that improve the adaptability and application of IDS models. Addressing these areas could promote DL and ML models used in IDS significantly, making them more powerful, more efficient, and more adaptable to the constantly evolving landscape of cyberthreats.

10 Discussion

This survey reviews current machine learning and deep learning techniques for intrusion detection systems, highlighting their strengths and challenges. Machine learning methods like Support Vector Machines and Random Forests work well for signature-based intrusion detection, using structured data to detect known threats [50]. However, they struggle with new attacks. Deep learning models, such as Convolutional Neural Networks and Long Short-Term Memory models, are better for anomaly-based detection, handling complex data like network traffic [115]. Combined models, like CNN-LSTM, improve detection of advanced attacks [13].

Challenges remain, as noted in the limitations section. False positives in anomaly-based systems create extra work for analysts [24]. Deep learning models can be slow for real-time use [24]. Graph Neural Networks are promising for analyzing network structures [124]. Explainable AI methods help make deep learning clearer [111]. Federated learning supports private detection, useful in healthcare [20]. These insights show the balance between detection accuracy and practical use, shaping better intrusion detection systems.

Author contributions

Abbas Mirzaei and Ali Hozouri conducted the literature review and structured the main body of the manuscript. Mehdi Effatparvar contributed to the analysis and categorization of IDS models. Abbas Mirzaei prepared the figures and comparative tables. Ali Hozouri critically revised and edited the manuscript. All authors reviewed and approved the final version of the manuscript.

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